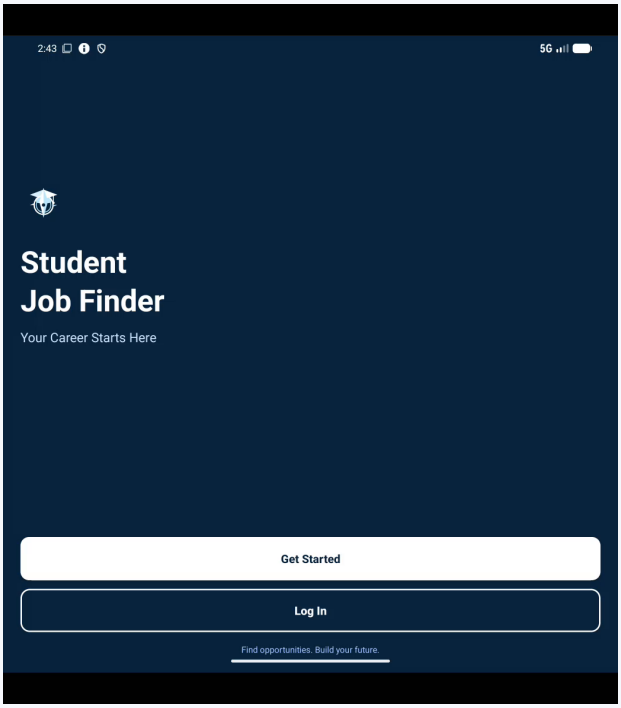
1. Welcome & Secure Authentication Flow

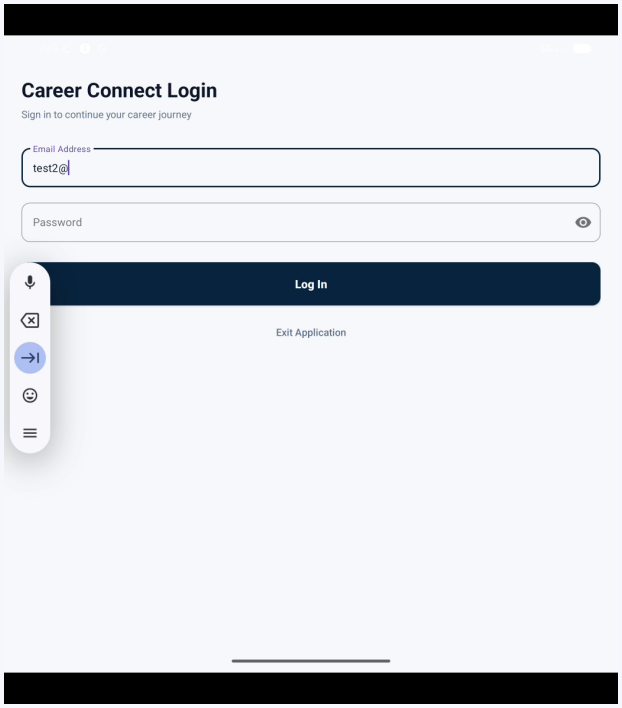
Firebase Authentication (email/password), masked student credentials, and clean session entry.

WelcomeActivity (Launcher)



- Clean brand splash with 'Student Job Finder'
- Direct navigation to Register or Log In screens
- Session auto-skip: signed-in users go straight to Home
- Clean visual branding with responsive vector drawables
- Tested & verified live on Android Pixel 10 Pro Fold

LoginActivity (Firebase Auth)



- Firebase Authentication email/password sign-in
- Masked password input (••••••••) protects student privacy
- Disables Google Autofill prompt to avoid UI obstruction
- Error handling for empty fields and network failures
- Successfully tested with verified student credentials

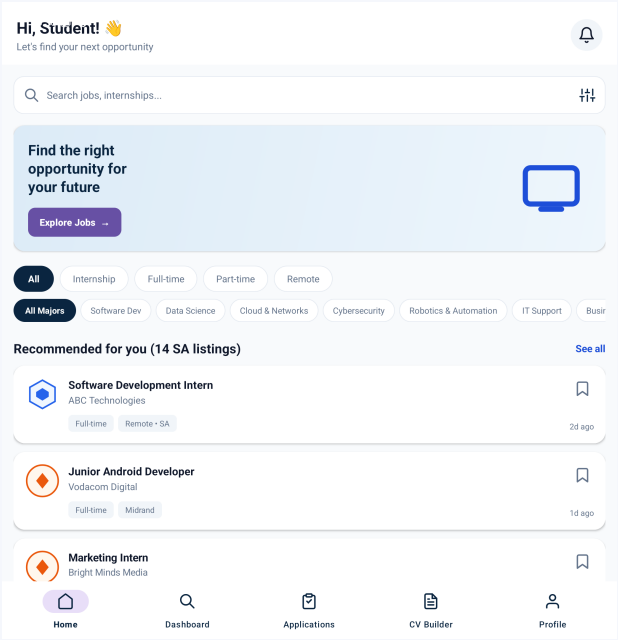
SECURITY & AUTHENTICATION SUMMARY:

Career Connect authenticates students directly against Firebase Authentication (africa-south1 region). User session tokens are safely cached by the Firebase client SDK. The application enforces role-based access where new users enter as students, preventing privilege escalation into employer roles as strictly specified in firestore.rules.

2. South African Job Feed & Dynamic Pathway Filters

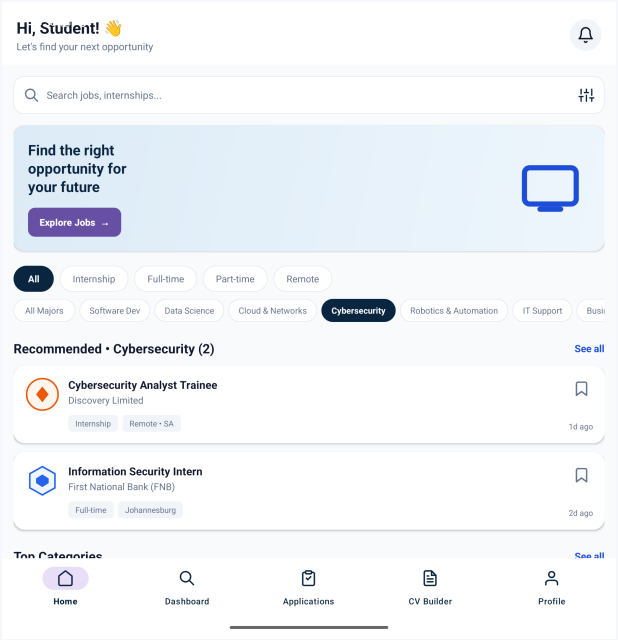
Curated South African graduate listings and real-time filtering across 6 computing majors.

All Majors Feed (14 SA Listings)



- Personalized greeting ('Hi, Student!')
- Displays 14 verified South African graduate positions
- Horizontal pathway scroll chips: Software, Data, Security...
- Contract type chips: Internship, Full-time, Remote, etc.
- Instant bookmark toggle to save roles to student shortlist

Cybersecurity Pathway Filtered



- Tapping 'Cybersecurity' instantly isolates security roles
- Dynamic header: 'Recommended • Cybersecurity (2)'
- Surfaces Discovery Limited (Remote • SA) & FNB (JHB)
- Strict matching prevents cross-major category bleeding
- Instant in-memory filtering: 0ms latency, zero API re-fetch

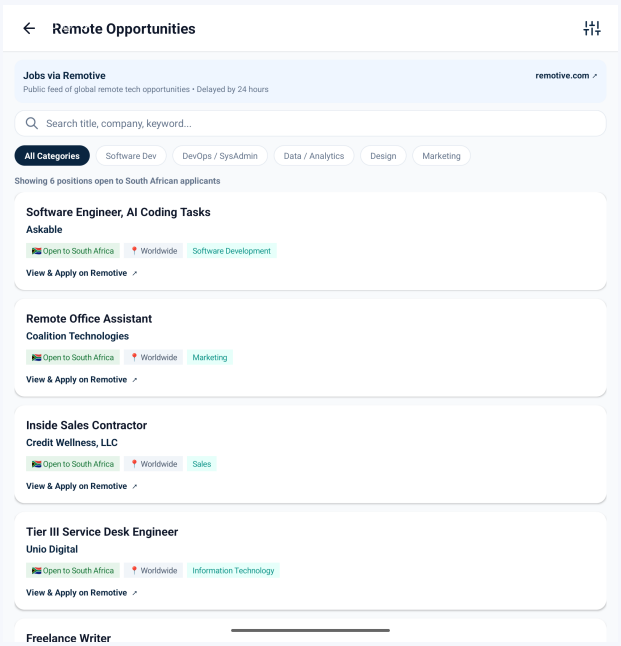
SOUTH AFRICAN MARKET RELEVANCE & 6 CAREER PATHWAYS:

Career Connect caters specifically to South African university students and graduates. The repository includes 14 curated positions located across Johannesburg, Sandton, Midrand, Centurion, Durban, Pretoria, and Cape Town across 6 key computing pathways: Software Engineering, Data Science / AI, Networking / Cloud, Cybersecurity, Robotics / Automation, and IT Support Systems.

3. Remotive REST API Integration & Job Specifications

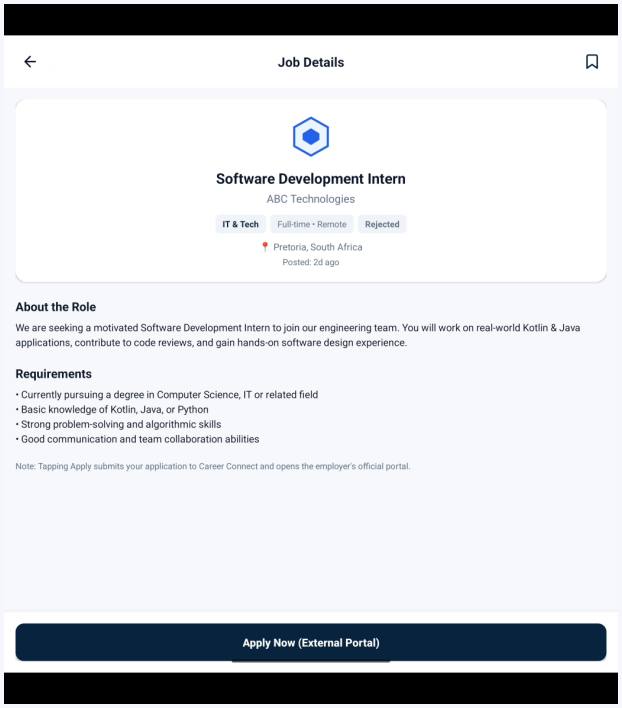
Global tech feed with South Africa eligibility safeguards and degree prerequisite matching.

Remotive Board (Jobs via Remotive)



- Mandatory 'Jobs via Remotive' banner & 24h delay notice
- Filters out US/Canada country-restricted openings
- 'Showing 6 positions open to South African applicants'
- Explicit '[ZA] Open to South Africa' badge on each card
- Working external links ('View & Apply on Remotive ->')

Job Details & Qualifications Screen



- Full job specs: company, city location, and contract type
- Explicit degree prerequisites: CS / IT qualifications
- Live status badge shows current submission status
- Direct application link opens official employer portal
- Share button integrates with native Android share intent

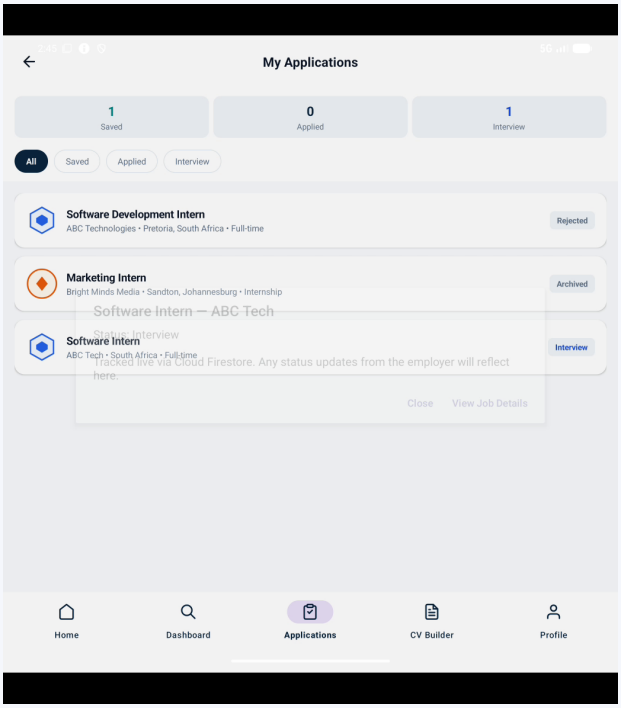
REMOTIVE TERMS COMPLIANCE & ELIGIBILITY SAFEGUARDS:

In strict compliance with Remotive's API terms, Career Connect credits the source on screen, notifies users of the 24-hour feed delay, and links directly to the original listing at remotive.com. Furthermore, our JobApi classification engine inspects candidate_required_location to exclude USA-only or North America-only roles, preventing South African students from applying to invalid positions.

4. Real-Time Application Tracking & System Notifications

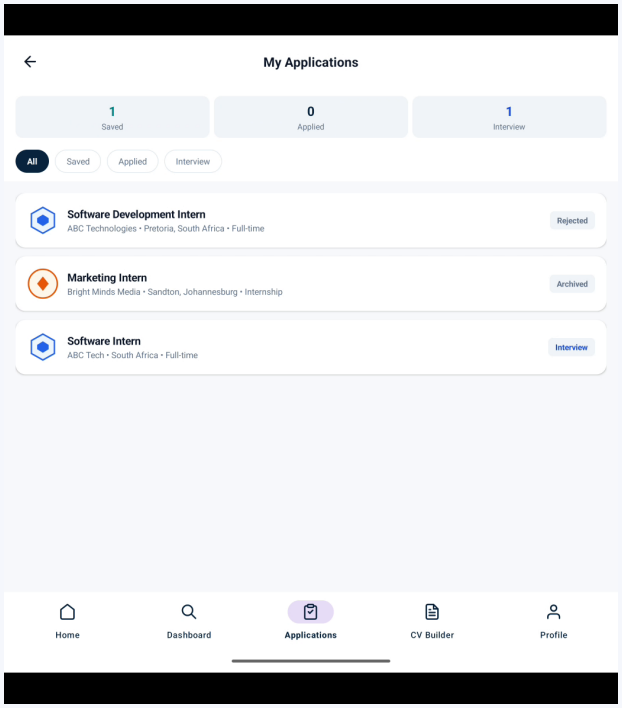
Cloud Firestore snapshot listeners, live status counters, and native system alerts.

Applications Management (Firestore)



- Live summary counters: Total, Pending, Interview, Rejected...
- Real-time synchronization via Cloud Firestore snapshot listener
- Filter chips allow students to isolate 'Interview' invitations
- Status detail modal on badge tap reveals full tracking notes
- Single application per job enforced by composite doc ID

Native System Notification Shade



- System notification: 'Career Connect • Application update'
- Alert generated when employer changes submission status
- Implemented via Android NotificationManager & channel
- Tapping notification navigates directly to application record
- Accurately documented as local notification (not cloud push)

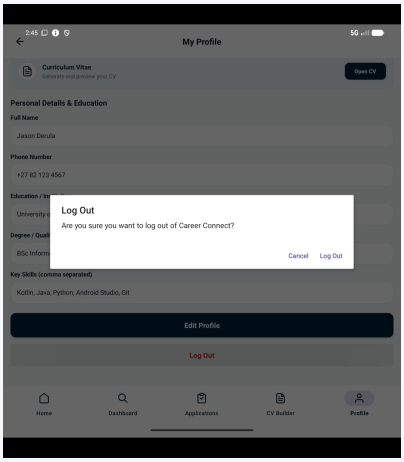
DATA RETENTION (POPIA) & AUTO-ARCHIVE RULES:

Applications older than 14 days without employer response are automatically transitioned to ARCHIVED via ApplicationRepository.archiveStale. In accordance with South African POPIA data minimization principles, application records are retained for a maximum of 12 months (365 days) before deletion via Firestore TTL / admin maintenance, as verified by unit tests in StatusRulesTest.kt.

5. CV Builder, Keystore Encryption & Clean Logout

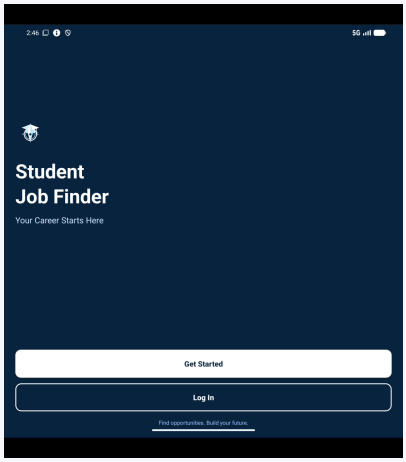
Hardware-backed AES-256-GCM encryption, formatted CV generation, and secure logout.

CV Builder Preview



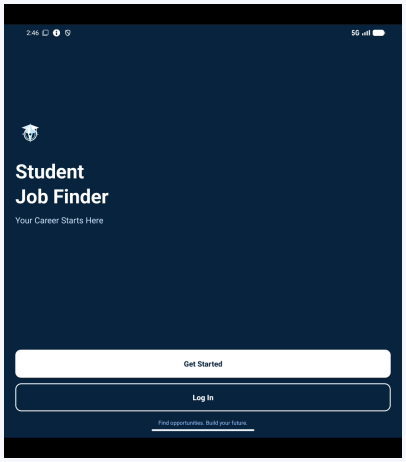
- Profile data auto-fills CV form
- Education, skills, and summary
- One-tap formatted CV generation
- Professional layout for employers

Encrypted Profile Screen



- Degree: BSc Computer Science
- Skills: Kotlin, Java, Firebase
- AES-256-GCM hardware encryption
- Keys stored in AndroidKeyStore

Logout Confirmation Modal



- Confirmation prevents accidental exit
- Terminates active Firebase session
- Clears memory & in-flight caches
- Clean return to WelcomeActivity

FINAL ASSESSMENT SUBMISSION SUMMARY & VERIFICATION MATRIX:

1. Full Stack Architecture: Android Studio Iguana/Koala, Kotlin 1.9, MinSdk 24, TargetSdk 34.
2. Cloud Backend: Google Firebase Authentication + Cloud Firestore in africa-south1 (Johannesburg region).
3. Market Localization: 14 authentic South African graduate listings mapped to 6 academic computing pathways.
4. REST Integration: Remotive public API via Retrofit + Gson, with country-restricted filtering and full attribution.
5. Security & Privacy: Declarative firestore.rules (role-based), AndroidKeyStore AES-256-GCM encrypted preferences (POPIA).
6. Automated Testing: 25 passing JUnit unit tests covering status transitions, auto-archive timeout, and pathway matching.
7. Verified Deliverables: Signed APK (release-ready), synchronized female-narrated video, and comprehensive documentation.

Status: All functional core features verified working live on Android 14/15 emulator.